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UNIMAS OPEN SCIENCE CONFERENCE

Universal Accessibility: Successful Open Science Models from Finland

From Data to Impact: Enriching Global Communities
21-22 October 2026

Co-funded by
the European Union

Dr. Erika Lilja | Tampere University

Tampereen yliopisto
Tampere University

Dr Erika Lilja | PhD in Science and Technology Studies and Administrative Sciences

RESEARCH SCOPE | Publications and research data were the main empirical focus because these were the most developed policy areas.



Science and Public Policy, 2020, 803–817
doi: 10.1093/scipol/scaa044
Advance Access Publication Date: 8 December 2020
Article

OXFORD

Threat of policy alienation: Exploring the implementation of Open Science policy in research practice

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Abstract

Many European countries have introduced Open
efficiency of science and to increase economic an
experiences of OS policies remain relatively unde
policy alienation perspective to understand resea
Analysis of survey responses indicates that resea
and that they feel policy alienation from OS policy
support OS policy implementation despite the fac
enhance the excellence and quality of research. T
lie not only in the research evaluation and acader
divide. These problems need to be solved in or
knowledge production of 'openness' itself.

Key words: Open Science; policy alienation; research policy; scie

1. Introduction

The Open Science (OS) movement has emerged as a charm
scientific progress and emphasized the ability to foster transp
equality, and innovation (Levin and Leonelli 2017). In Euro
example, OS is currently being promoted by the Eu
Commission, Member State governments, European resear
ders, various research organizations, and learned societies.
some time, when OS picks up momentum amongst policy
and openness is seen as something that needs to be system
promoted, the assumptions and values embedded in OS poli
main unclear.

Science has been based on open knowledge production an
ng of research findings for many centuries (Merton 1973; Di
and David 1994; David 2008). OS today is based on the id
scientific knowledge, which is shared and developed through
erative networks, should be transparent and accessible (V
Sanz and Martínez-Fuentes 2018). OS policies—understood
justification, management, prioritization, and funding of basi
ific research based on the advancement of OS—define st
search outputs and practices more valuable than others an
assumptions about what, who, when, and how openness
occur (Whyte and Pryor 2011). Whereas OS seems to encom
most any discussions about the future of scientific kno

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Science and Public Policy, 2025, 00, 1–13
DOI: https://doi.org/10.1093/scipol/scaf021
Article

OXF

Responsibility dynamics in open science: exploring researchers' framing of ethical considerations in policy implementation

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This study explores the responsibility framings researchers adopt when implementing Open Science (OSI policies, focusing on how
bility dynamics are integrated into research practice. Three key responsibility framings emerged: internal orientation to responsibility (
ethics), external orientation to socially/societally responsible science (social and societal ethics), and external orientation to shared,
responsibility (collective ethics). These framings influence OS policy implementation, highlighting researchers' roles in navigating att
lenges and shaping OS practices. The first framing emphasizes personal ethics and reflection on OS policies, the second stresses th
impact of research, and the third underscores the need for collective governance. This study contributes to understanding responsibi
ance by presenting a dynamic, proactive view of responsibility, where researchers actively engage, challenge, and shape policies. It off
recommendations for adaptive OS frameworks, critical reflection on ethical challenges, and inclusive governance involving diverse sta
to foster responsible Open Science.

Keywords: Open Science; open access; data sharing; responsibility; science policy; governance; ethics.

1. Introduction

In the past decades, Open Science (OS) has broadened into dis-
cussions about the future of knowledge creation and scholarly
communication (Fecher and Friesike 2013) and is recognized
as a key policy priority and an instrument for governance
by the European Commission (EC), a standard in research
and innovation funding. Commitment to OS in science policy
arises from its potential to improve research practices' qual-
ity, efficiency, and responsiveness. According to the European
Commission (2023), sharing knowledge and data early with
relevant actors and involving partners from academia, indus-
try, public authorities, and citizen groups enhance creativity
and trust in science.

In Europe, OS policies are closely linked to the debate on
the social responsibility of science, shaped by Responsible
Research and Innovation (RRI). Both aim to make research
more inclusive, transparent, and aligned with societal needs.
RRI seeks to democratize research through public engagement
and cross-sector collaboration, while OS advocates for trans-
parency, accessibility, and knowledge sharing, encouraging
collaboration between researchers, policymakers, and societal
actors. Both frameworks emphasize ethical research practices
and the societal impact of research, promoting responsible
approaches to meet current and future needs. The EC's con-
cept of OS can be seen as a continuation of its promotion
of the RRI programme, which supports the goals of OS by
ensuring that research is not only open and accessible
conducted with a sense of responsibility towards socie
being and ethical considerations. The EC's role as a
in multi-level governance and dynamics (Fisher and R
has been crucial in integrating both RRI and OS into
frameworks across Europe.

Formalized openness in policies establishes ope
distinctions, emphasizing certain outputs and pra
more valuable than others (Levin and Leonelli 201
reinforces normative views on the economic and socie
significance of scientific research and outcomes. Previo
ies show that researchers' openness decisions are sh
institutional, policy, and technical factors (Levin et a
and that in the European and national policy lan
numerous OS initiatives have caused complexity an
sion (Chataway, Parks, and Smith 2017). Levin and
(2017) argue viewing openness as a performative and
tive process. Their understanding emphasizes context
factors, aligned with Haeussler et al. (2009), and ave
plistic OS definitions and the idea of sole openness
societal issues typically embraced in policy. Given
of consensus on OS, researchers often grapple with
tainty in implementing these policies (Bozeman 2012
Rolando, and Bozeman 2013; Ferguson 2014; Grand,
son, and Bultitude et al. 2016; Chataway, Parks, an
2017; Lilja 2020). Diverse policies with varying co

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Science and Public Policy, 2026, 00, 1–13
DOI: https://doi.org/10.1093/scipol/scag012
Published: 28 July 2026
Article

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Why do Open Science policies fall short? Institutional logics and ethical framings in policy implementation in Finland

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This article explores the ethical and institutional dimensions of Open Science (OS) policy implementation in hybrid research organizations in Finland by mapping ethical framings across five institutional logics. It uncovers reinforcing ethical framings across institutional logics that support OS implementation; however, it also reveals tensions, especially where market and corporate logics conflict with professional and disciplinary values and norms. Drawing on both survey and interview data and by combining the institutional logics framework with the concept of openness as a performative process, the study offers a novel analytical lens to understand how researchers interpret and negotiate OS policies in practice. It contributes to ongoing scholarly discussions on science policy implementation and dynamics that shape OS practices and by proposing a typology of researcher responses ranging from advocacy to policy alienation. These responses span a continuum from proactive engagement to ethical resistance and disconnection.

Keywords Open Science, institutional logics, openness, ethics, science policy

1. Introduction

This article examines the implementation of Open Science (OS) policies in Finland within hybrid organizations that integrate traditional academic values with contemporary state and market-driven demands. As OS aims to transform paradigms, it offers both opportunities and challenges, especially in public universities and research organizations that have become more complex in the last twenty years.

The evolving dynamics of public higher education are influenced by neo-liberal reforms, including New Public Management (NPM) and managerialism, resulting in tensions between market-driven pressures and academic values that emphasize autonomy and the public good (Pekkola and Kivistö 2016; Bleiklie et al. 2017; Hazelkorn et al. 2018; Grossi et al. 2019; Pekkola et al. 2018). The increasing complexity of public higher education systems is driven by external demands for societal relevance, accountability, and performance (Kallio and Kallio 2014, Kallio et al. 2016, Hazelkorn et al. 2018, Kivistö et al. 2019), and in the evolving science policy environment, universities and public organizations have transformed into hybrid entities, integrating commercial principles with conventional academic values (Battilana and Lee 2014, Denis et al. 2015, Gebreiter and Hidayah 2019, Pekkola et al. 2020, 2022). Recent studies have highlighted the complexity and hybridity of science policy. This stems from tensions between different institutional logics and the need to balance numerous interests and demands. Thus, the dynamic between state logic and market logic has emerged as a recurring theme in research, with studies identifying tensions and adaptive strategies aimed at balancing these logics in policy implementation. (*E.g.* Lattu and Cai 2023; Turner et al. 2023)

Recent work shows that emerging digital technologies, expanding open infrastructures, and evolving research practices are reshaping the role of universities and researchers, calling for new normative structures, governance models, and forms of open scientific work (Vicente-Saez et al. 2021). Given that OS policies promote transparency, accessibility, and sharing (*e.g.* Vicente-Saez and Martinez-Fuentes 2018), they may conflict with hybrid organizations' institutional logics, necessitating an understanding of how these tensions arise at the micro-level of research practice. Moreover, since OS policies aim to influence cultural transformation in research (see *e.g.* LERU 2018, Open Science Coordination in Finland 2025), it is important to examine how hybrid organizations entangled with state and market logics may either promote or hinder openness in science.

In this article, Open Science refers to a set of contemporary policy initiatives and practices aimed at institutionalizing openness in research. While it draws on the broader tradition of openness, OS is characterized by specific goals, such as open access publishing, open data, and public engagement. OS initiatives frequently presume a single definition of openness and ignore the complexities involved in different situations (Whyte and Pryor 2011). Openness, as discussed in the sociology and philosophy of science (*e.g.* Merton 1973, Latour and Woolgar 1979), however, refers broadly to the ethos of transparency, accessibility, and the sharing of knowledge. It encompasses a range of practices and values, from data

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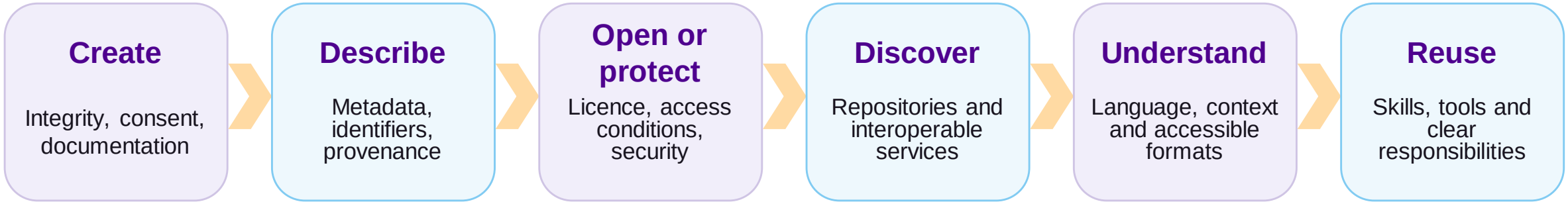
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Join the journey

1. The Finnish model
2. What success looks like
3. What research reveals
4. Where to go next

From accessibility as access to accessibility as capability

Accessibility is an end-to-end pathway



A break at any stage creates “shallow accessibility”: technically open, practically unusable or ethically unsound.

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Finland’s model is an ecosystem, not a policy programme.



Shared vision

National declaration and four policy areas

Co-creation

Universities, institutes, funders, libraries and researchers

Shared services

Repositories, identifiers, discovery and preservation

Incentives

Funding, assessment and organisational commitments

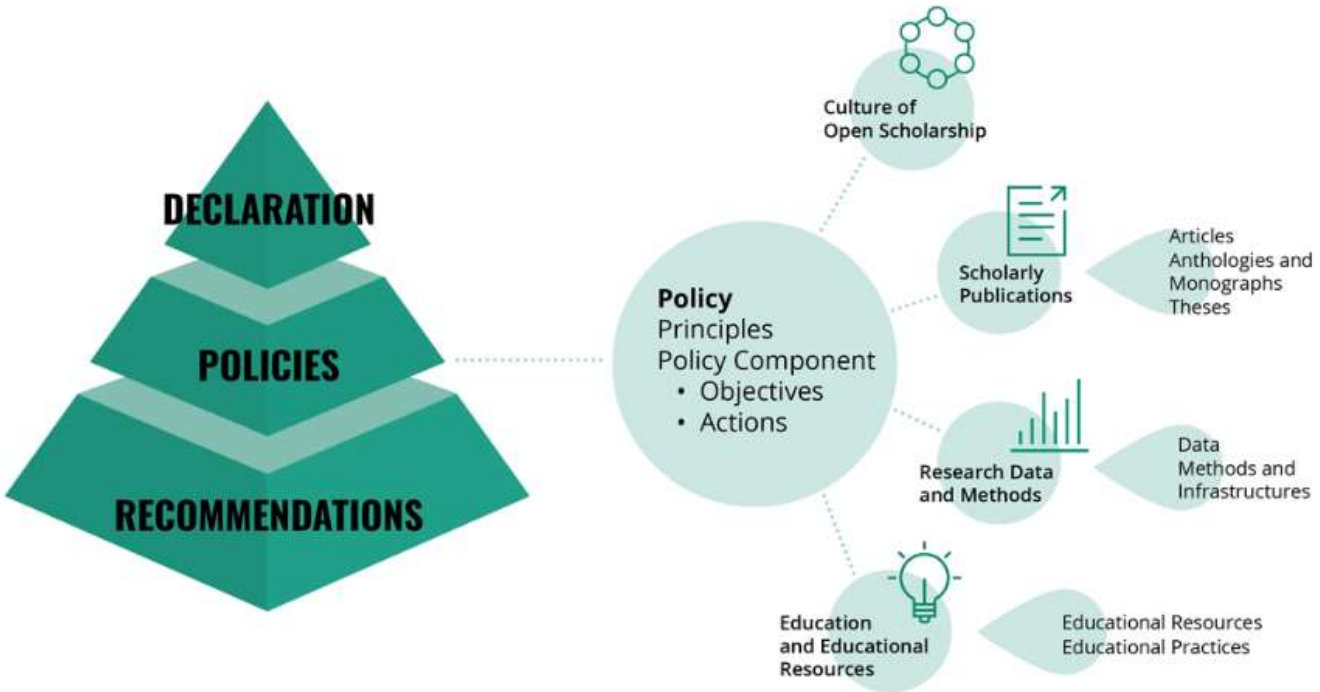
Support

Guidance, training and professional expertise

No element works alone. Coordination turns separate initiatives into a durable national capability.

OPEN SCIENCE AND RESEARCH POLICIES

Last updated 5.12.2023



Open scholarship is a challenge and opportunity for the whole research community

Declaration for Open Science and Research

Policies of open science and research in Finland

Recommendations of open science and research in Finland

Reference Architecture 2024–2030

Monitoring

Funding Open Science

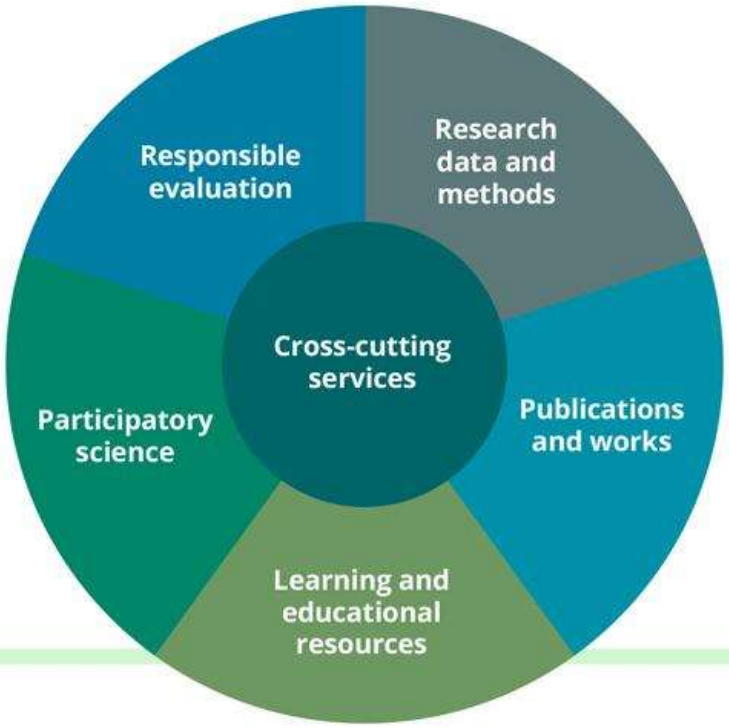
International statements

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Open science and research reference architecture 2024–2030 in Finland

c. Service map

The diagram (p. 86–87) illustrates the key service areas for the reference architecture. The service areas have been further divided into services and service processes, which are not shown in the diagram. The cross-cutting services are positioned in the middle of the other service areas.



WHAT DID WE LEARN?

The large number of **CROSS-CUTTING SERVICES** in relation to other service areas shows that the individual service areas of open science and research have a great deal in common.

LEGAL AFFAIRS are a key aspect of the promotion of open science and research for two reasons: the promotion of open science requires competence with openness-related legislation and legislative influencing in connection with the promotion. At a national level, the status of both remains poor.

The challenge in **RESPONSIBLE EVALUATION** is that inappropriate indicators are still being used, as are inappropriate methods in using more responsible indicators. To advance high-quality evaluation processes, support is required for both the evaluation methods and the generation of a knowledge base.

OPEN EDUCATION SERVICES enable research-generated knowledge being disseminated into society. In addition, open education allows learners to be guided onto the path of open science from an early stage onwards.

Established practices exist in **OPEN SCHOLARLY PUBLISHING**, but the sector is continuously developing, and continued high quality must be ensured. Open scholarly publishing may seem challenging to researchers and organisations, which is why it is important to invest in support services and economically viable publishing models.

At the local and national level, in particular, little practical support and few tools are available for **PARTICIPATORY SCIENCE**, even though e.g. citizen science is attracting interest and occasionally gaining media visibility.

THE RESEARCH DATA AND METHOD SERVICES are still in development. Only computing services are currently adequately available at the local, national and international level. Making research datasets and methods openly available requires systematic and determined efforts to achieve the objectives of 2030, determined in the reference architecture.

Open Science and Research Reference Architecture 2024–2030 | Open Science

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Finland in the international Open Science landscape

HARD MANDATE

Finland belongs to the smaller group of European countries with relatively broad and binding national Open Science policies.

FRANCE • IRELAND • SLOVENIA • FINLAND

SOFT MANDATE

In many European systems, national policies rely more strongly on recommendations and voluntary implementation.

BELGIUM • DENMARK • GERMANY • NETHERLANDS • SPAIN

POLICY DEVELOPMENT

Elsewhere, Open Science often begins with publications and research data before expanding towards a wider policy framework.

GLOBAL PATTERN: OA + DATA → BROADER OPEN SCIENCE

FINLAND

Strong national co-ordination
A shared goal across the research community

Broad policy scope
Beyond publishing towards data, culture, education and assessment

Community co-creation
Policies and implementation developed through national expert networks

Finland is a forerunner, but not a universal template or model.
National strengths still need context-sensitive implementation in research practice.

Lilja (2026); Rochambeau & Konach (2022); UNESCO (2023).

The persistent gap: policy ideals and research realities

POLICY IDEALS

- Transparency
- Accessibility
- Efficiency
- Societal impact

MIND THE IMPLEMENTATION GAP

RESEARCH REALITIES

- Disciplinary diversity
- Ethical dilemmas
- Competing incentives
- Time and resource constraints

Researchers might support openness in principle but question how particular mandates work in practice.

What has worked well?

1. National co-ordination

A shared declaration, policy architecture and community-led development create continuity beyond individual projects.

2. Shared infrastructures

National and institutional services lower the threshold for managing, publishing, finding and preserving outputs.

3. Research support

Libraries, data stewards and research services translate principles into workable choices and local guidance.

4. Responsible assessment

A broader understanding of outputs and contributions can align recognition with open practices.

5. Collective learning

Monitoring, pilots and peer exchange allow models to mature while revealing unintended effects.

These features do not remove tensions, but they create capacity to identify and address them.

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Responsibility in Open Science framed in three ways by researchers affiliated to Finnish research organisations

INTERNAL

Personal integrity and professional standards

"What is the responsible choice in this research context?"

SOCIETAL

Relevance, interaction and implications beyond academia

"How does openness serve people and society?"

COLLECTIVE

Shared duties across researchers, organisations, funders and policymakers

"Who must act together for openness to be responsible?"

Responsible openness is not only an individual obligation.
It is a governance relationship and a collective responsibility.

Ethical framings: the same policy can mean different things

Openness as a duty

- Publicly funded knowledge should benefit society
- Access should be equitable
- Transparency supports public trust

Openness as integrity

- Transparency can improve quality and reproducibility
- Responsible sharing requires safeguards
- Context and disciplinary standards matter

Openness under pressure

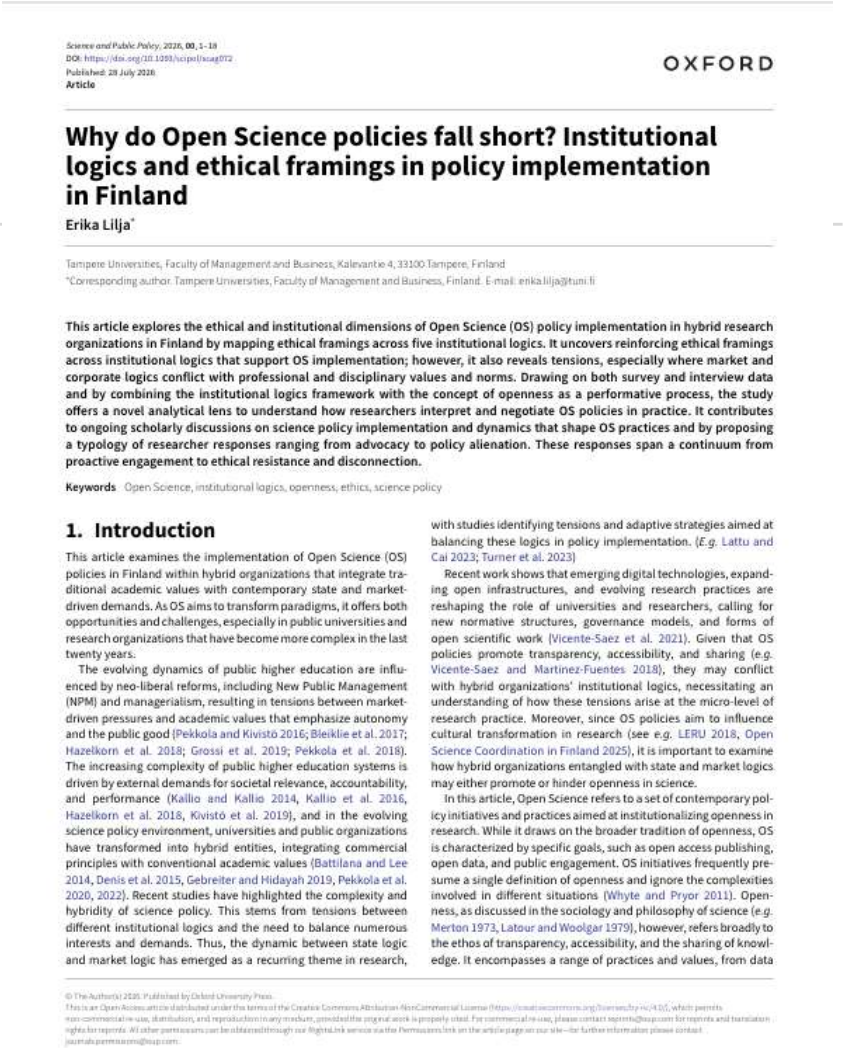
- APCs and commercial control can reproduce inequality
- Metrics can reward visibility over collaboration
- Mandates may produce shallow compliance

Policy efficiency depends on whether these ethical concerns are heard, balanced and supported.

Why do Open Science policies fall short?

Studying OS implementation in research practice within hybrid research organisations.

- Researchers interpret policy through multiple values and obligations.
- Professional and disciplinary norms may reinforce openness, but also set legitimate limits.
- Market and corporate pressures can conflict with public-good and integrity-based goals.
- Implementation is negotiated in everyday research, not mechanically adopted.



Science and Public Policy, 2026, 00, 1–18
DOI: https://doi.org/10.1093/scipol/scag072
Published: 28 July 2026
Article

OXFORD

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Institutional logics can reinforce or undermine openness in research practice

REINFORCING:
aligned values make openness meaningful

CONFLICTING:
misaligned demands produce resistance and obstacles

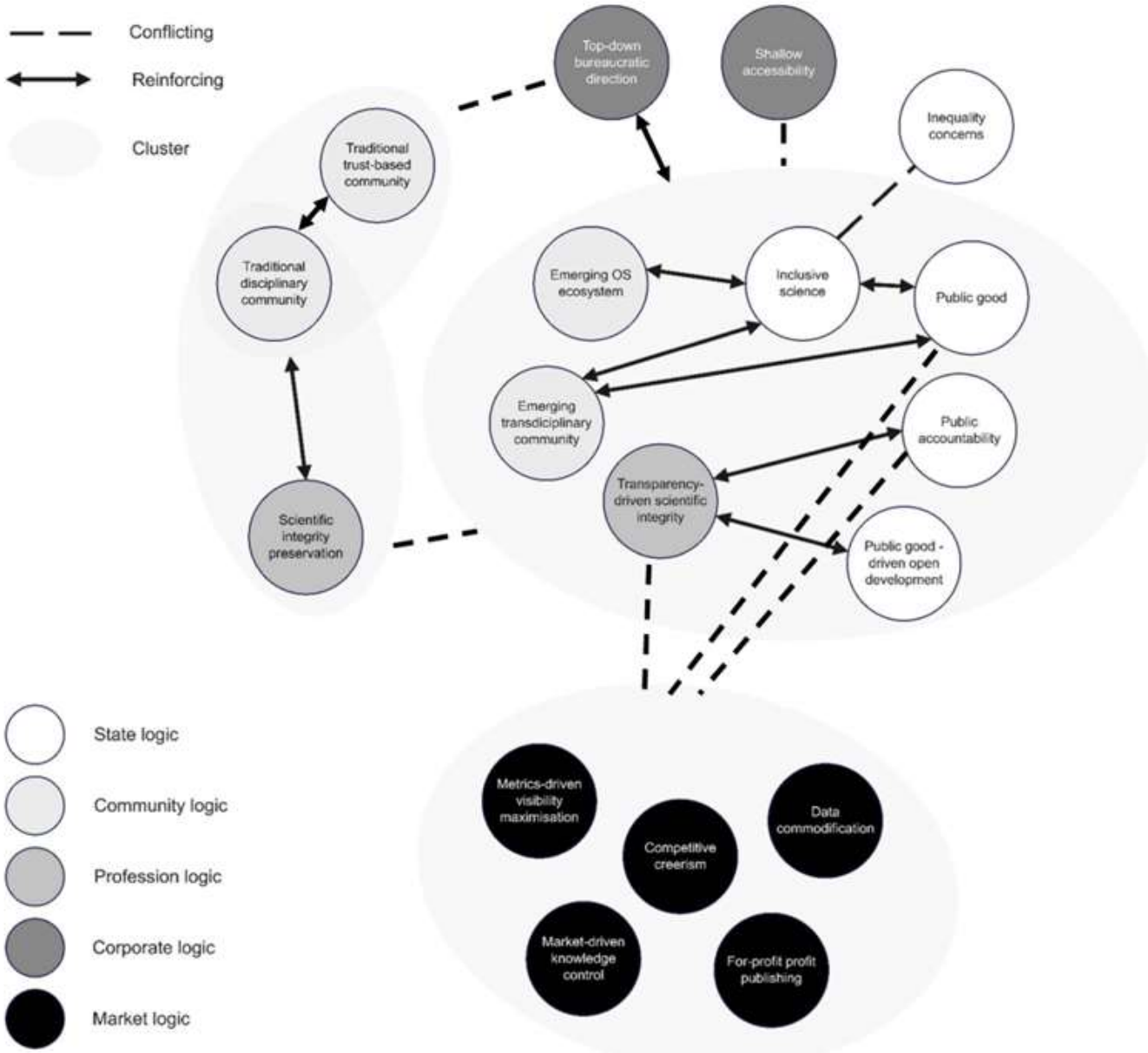


Figure 1 Reinforcing and conflicting ethical framings and clusters across institutional logics.

Science and Public Policy, 2026, Vol. 00, Issue 00

Researchers responses extend beyond compliance or resistance.

These are fluid positions, not fixed researcher types. Researchers may move between them as context changes.

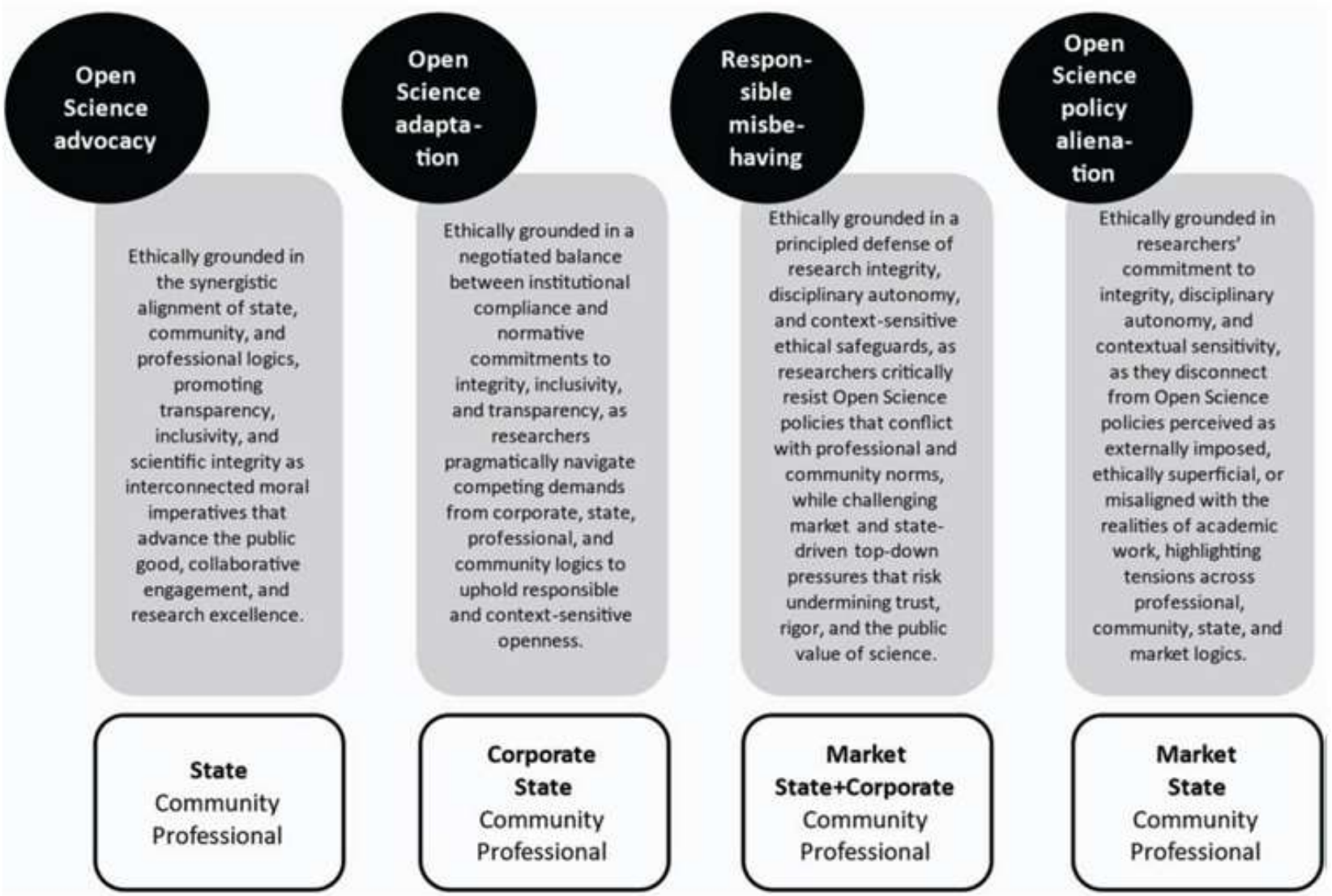


Figure 2 Four response types and their institutional grounding in OS policy implementation.

The Finnish agenda is shifting towards responsible openness

Declaration for Open Science and Research 2025–2030



Areas of the new Declaration. New addition is a section for guiding principles for promoting openness. Photo: Declaration for Open Science and Research 2025–2030.

New emphasis: geopolitical change, research security and rapidly evolving technologies must be considered without abandoning trust and international collaboration.

Rights Retention Strategy: access without surrendering author rights

1

University grants a prior licence

A policy secures rights in the author-accepted manuscript.

2

Researcher submits an article to a journal

Choice of publication channel remains with the researcher/author.

3

AAM is deposited

The peer-reviewed manuscript is made openly available.

4

Access is preserved

Immediate availability and long-term institutional preservation.

Finnish universities are introducing a shared prior-licence model during 2026.
The model supports immediate self-archiving, academic freedom and resilient access to research.

2.5 min

maDMPs: from static documents to living infrastructure

TRADITIONAL DMP

A largely free-text plan

Read, reviewed and updated manually

Often isolated from other systems

MACHINE-ACTIONABLE DMP

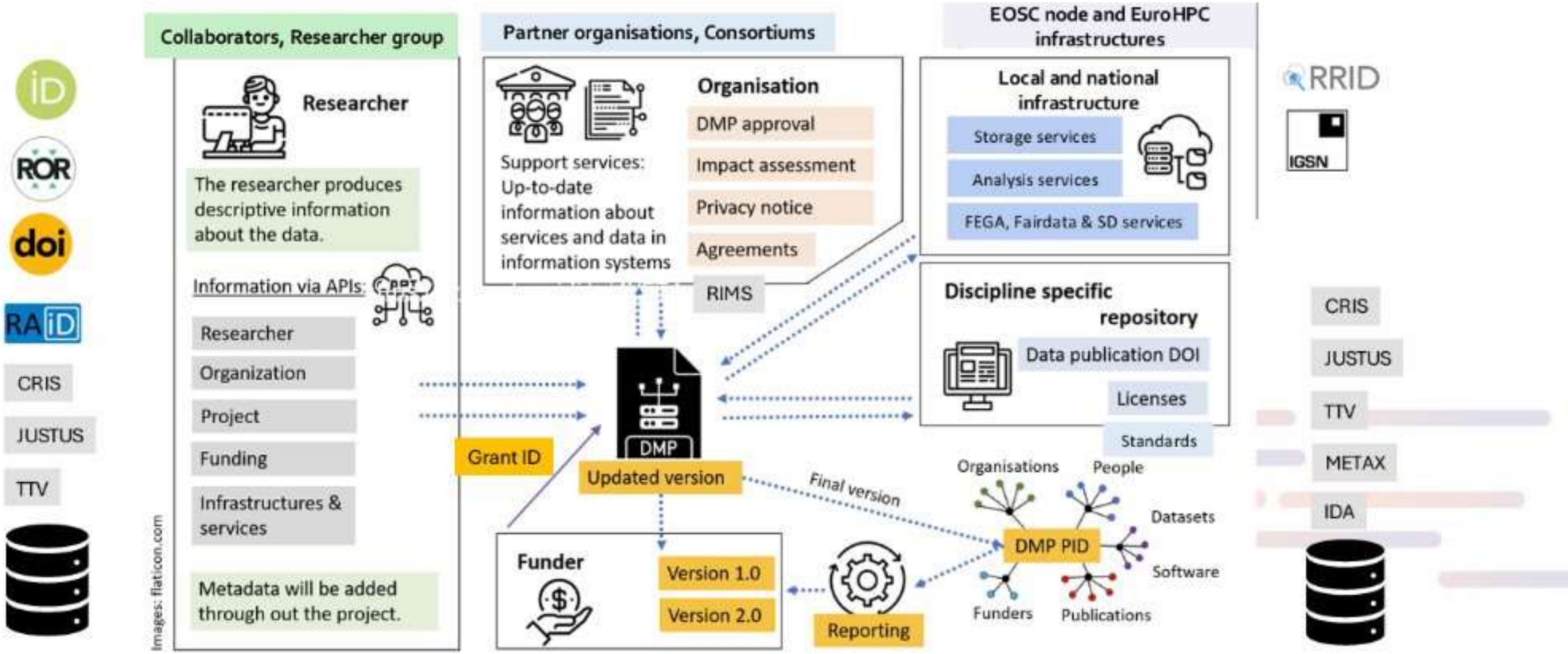
Structured and semantically enriched

Connects people, organisations, data, costs, licences and repositories

Reusable across research systems

Potential benefits:
less duplicate data entry • better planning • automated exchange • up-to-date oversight • FAIRer workflows

maDMP core idea in research ecosystem – interoperable data structures enable data flows serving the researcher



Artificial intelligence changes both access and responsibility



AI can expand access

Translation, summarisation, metadata enrichment, discovery and new interfaces may help more people and machines use research.

AI can distort access

Synthetic content, opaque models, bias, hallucinations and automated extraction can weaken provenance and trust.

Open science response

Document AI use; protect confidential or sensitive material; preserve provenance; clarify permissions for training and reuse; maintain human accountability.

2 min

Research security: “as open as possible” needs a new risk lens

WHAT?

Could the knowledge, data, code or infrastructure create misuse or dependency risks?

WHO?

Who needs access, and are roles, affiliations and responsibilities clear?

WHEN?

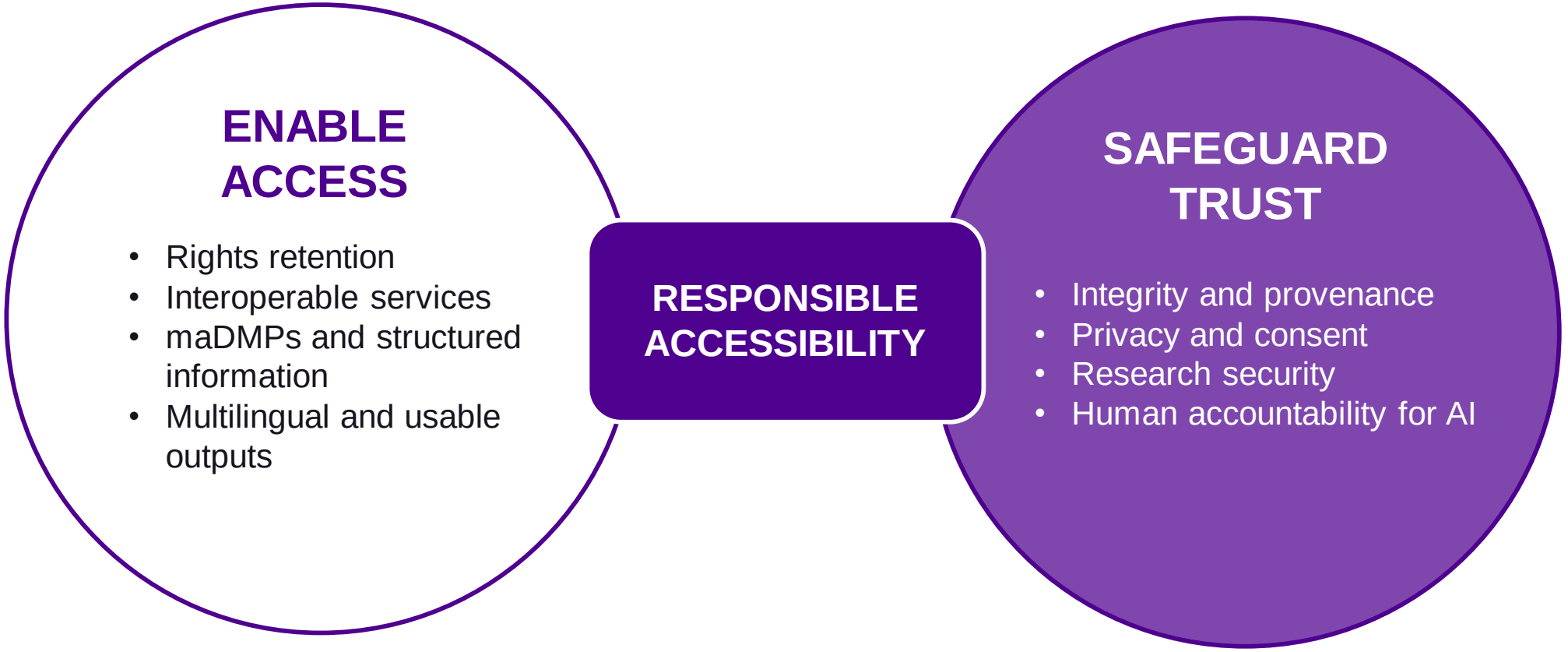
Can access be staged, delayed or reviewed as risks change?

HOW?

Which technical, contractual and governance controls are proportionate?

Responsible openness is not secrecy by default.
It is explicit, proportionate and revisable decision-making.

The emerging Finnish model: responsible accessibility



What Finland’s experience teaches

1. **Co-ordinate nationally** but implement with local and disciplinary sensitivity.
2. **Invest in shared capacity** because mandates without services transfer costs to researchers.
3. **Reward responsible practice** not merely output counts, visibility or formal compliance.
4. **Value researcher agency** through participation, clarity and room for justified exceptions.
5. **Evaluate meaningful access** including usability, equity, integrity and long-term availability.

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Thank you!

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